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MGU-BVOC (HONOURS)REGULAR EXAMINATION APRIL 2026

SECOND SEMESTER

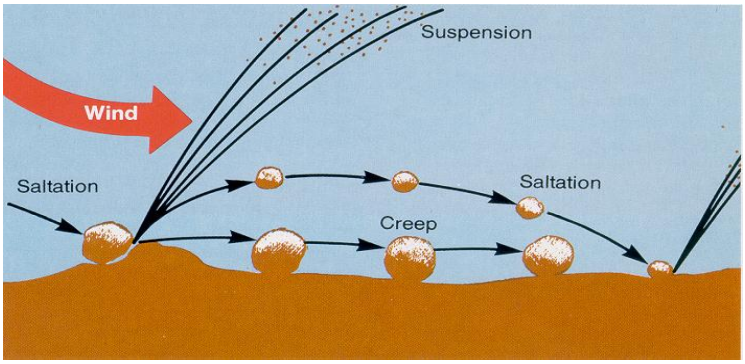
SKILL DEVELOPMENT COMPONENTS (SDC)

MG2SDCBAT102-FUNDAMENTALS OF SOIL SCIENCE

(2025 Admission Onwards)

PART -A

1.

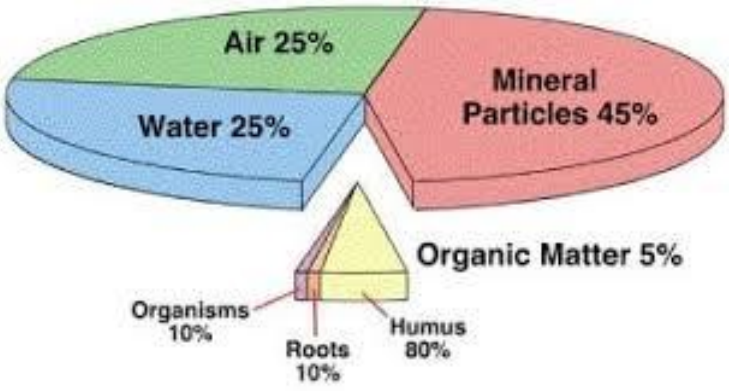


2. Soil physical properties

Texture, structure, density, porosity, colour, consistency ,water and air

3. It is completely weathered.Surface soil is porous.Aeration status of surface soil is good.The number and activity of soil micro-organisms is very high.Higher organic matter content.Due to presence of high organic matter content the colour of surface soil is deep brown or dark.It is fertile.Most of the essential plant nutrients are present.Surface soil has no hard pan.Cation exchange capacity is very high.It has good physical condition

4.



5. Hue-it denotes the dominant spectral colour (red, yellow, blue and green).

6. Essential plant nutrients are categorized as macronutrients (required in large amounts) and micronutrients (required in small amounts), with the 17 essential elements including carbon, hydrogen, oxygen, nitrogen, phosphorus, and potassium, among others.

7. Functions of Organic Matter:

- Physical: Improves soil structure, reducing compaction, and enhancing aeration and water infiltration.

- Chemical: Increases Cation Exchange Capacity (CEC), allowing soil to store nutrients (like N, P, K) and improving nutrient availability to plants.
- Biological: Acts as the primary energy source and habitat for soil organisms (bacteria, fungi, earthworms)
- Environmental: Enhances carbon storage (sequestration), which mitigates climate change and helps build resilient soil structure that reduces erosion by water and wind.

Benefits to Soil Health:

- Water Management: Acts like a sponge, holding water, which increases drought resistance.
- Soil Nutrient Management: Provides a slow-release, steady supply of nutrients, reducing the need for heavy chemical fertilizer application.
- Soil Structure: Darkens soil, increases aeration and reduces erosion.

PART -B

- **8. The Purpose of Soil Testing**
  - Soil testing helps in estimation of available plant nutrients.
  - It helps to group soils into classes related to the levels of nutrients
  - It helps in predicting yield
  - It helps to evaluate soil productivity and fertility
  - It helps to determine soil conditions like acidity, alkalinity and salinity.
  - It informs the farmers about the nutrient deficiency in a particular soil

9. Organic manure is a natural fertilizer derived from decaying plant and animal waste, used to enrich soil fertility and provide nutrients to plants. It is made from sources like animal dung, crop residues, and kitchen waste, and it improves soil structure, aeration, and water-holding capacity.

-Classification of organic manures

Bulky organic manure

Bulky organic manure refers to materials of plant and animal origin that are applied in large quantities due to their low nutrient concentration, but high organic matter content. Common examples include farmyard manure (FYM), compost, and green manure. These manures improve soil structure, water retention, and nutrient availability.

- **Farmyard manure (FYM)**: A mixture of cow dung, urine, and other organic matter, usually collected and stored in a roofed enclosure.
- **Compost**: Decomposed organic materials like yard waste, food scraps, and other plant-based matter.

- **Green manure**: Plants grown specifically to be incorporated into the soil while still green to add organic matter and nutrients.
- **Other types**: Sewage sludge and seaweed can also be considered bulky organic manures, though they may require special treatment before use to remove harmful components like pathogens and heavy metals.

Concentrated organic manure

Concentrated organic manure is a type of nutrient-rich fertilizer derived from animal and plant sources, containing a higher percentage of plant nutrients like nitrogen, phosphorus, and potassium compared to bulky organic manures. Examples include oil cakes, blood meal, bone meal, and fish manure.

- **Oil cakes**: The by product of extracting oil from seeds, such as groundnut cake and neem cake.
- **Blood meal**: Dried and powdered animal blood.
- **Bone meal**: Crushed animal bones, rich in phosphorus and calcium.
- **Fish manure**: Processed fish waste, also known as fish emulsion or fish meal.
- **Bird guano**: Dried droppings from birds or bats.
- **Horn and hoof meal**: Ground horns and hooves of animals.

10. Deficiency symptoms of calcium

- localized rot of tissues,
- leaf tip burns and root tip burn.
- Deformed leaves: New leaves are often small, curled.
- Yellowing: Random yellow spots can appear on the leaves.

Deficiency of magnesium

- Interveinal chlorosis (yellowing between the veins)
- Stunted growth,
- Reddish spots
- Necrosis (tissue death)

sulphur deficiency

- Yellowing of young (top) leaves, known as chlorosis, is a primary symptom.
- Stunted growth.
- Reddening of stems and petioles in some plants.
- Reduced yield and quality.

11. SOIL STRUCTURE

The arrangement of primary particles and their aggregates into a certain definite pattern is called soil structure.

1. Types of Structure

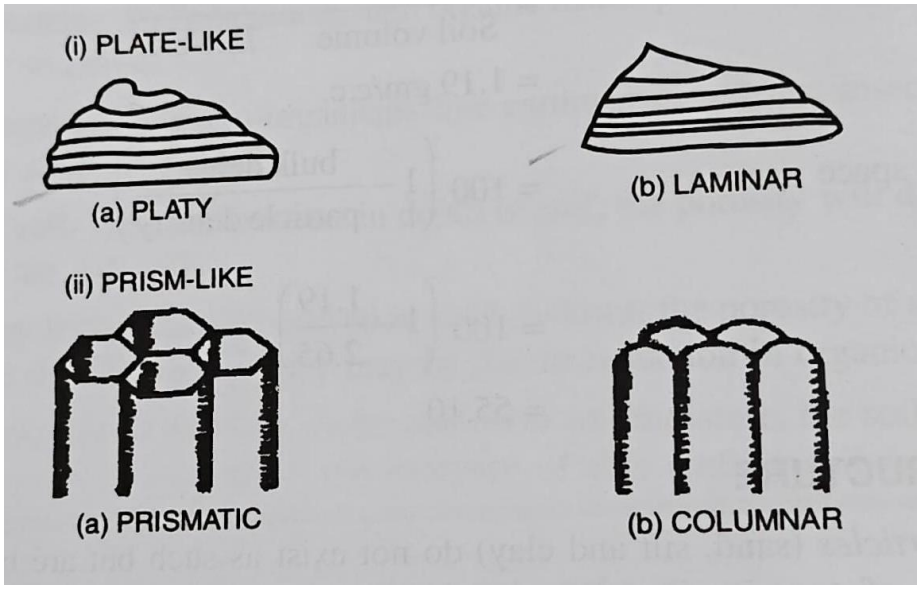
(i) Plate like,(ii) Prism like,(iii) Block like (iv) Spheroidal.

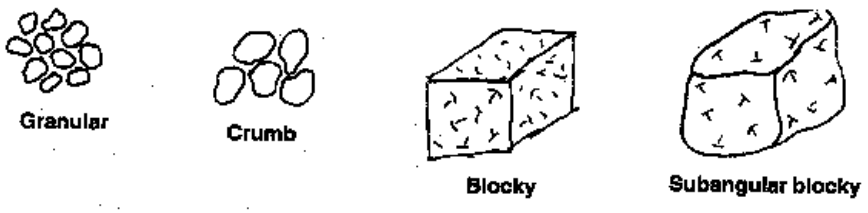
Plate like: The horizontal dimensions resulting a flattened, compressed When the units thick, they are called platy and when the units are thin, they are called laminar.

Prism like: The vertical axis, with flattened sides, giving a pillar-like shape. It has also two sub-types: columnar when the top is rounded, and prismatic when the tops of the prisms are still plane, level and clean cut.

Block like: Block like structure has also two sub-types: angular blocky when the faces are flat and edges of the cubes are sharp angular, and sub-angular blocky when the faces and edges are mainly rounded.

Spheroidal (sphere like irregular faces. Spheroidal type of soil structure has two structural sub-types: granular, it is less porous, and crumby when the granules are especially porous.





12. Particle density.

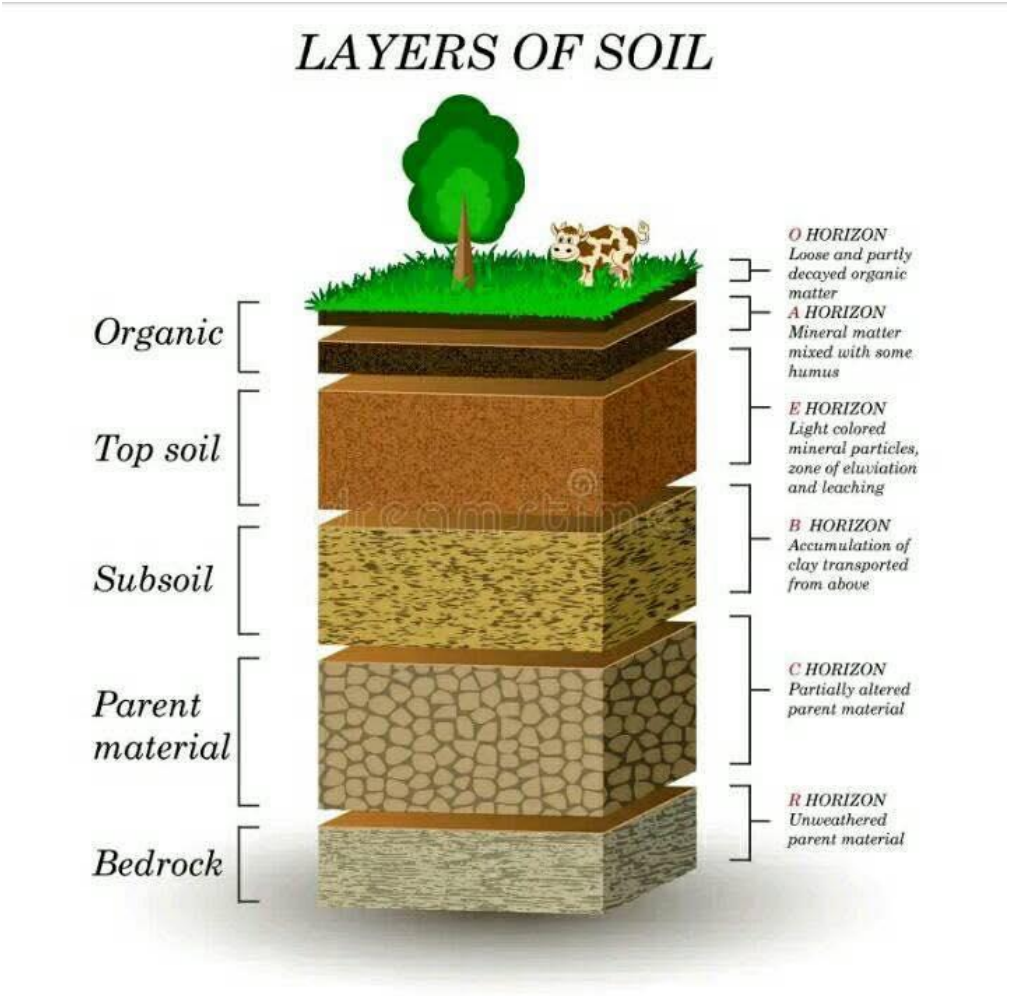
The weight per unit volume of the solid portion of soil is called particle density. It is also termed as true density. It is expressed in gm/c.c. Generally, in the normal soils the particle density is 2.65 grams per cubic centimetre

❖ **Bulk density.**

Bulk density is defined as the mass (weight) per unit volume of a dry soil It is also expressed in gm/c.c. The bulk density of a soil is always smaller than its particle density. Generally, in normal soils bulk density ranges from 1-1.33gm/c.c

13. SOIL PROFILE

A soil profile is a vertical cross-section of the soil that reveals its various layers, or horizons. These layers, such as the topsoil (A horizon), subsoil (B horizon), and parent material (C horizon), have distinct characteristics in color, texture, and chemical composition that are formed by soil-forming processes. A soil profile is used to understand the soil's properties for agriculture and conservation.



The layers of soil

Mature soil is made up of different horizontal layers, called horizons, which are visible in a vertical cross-section called a soil profile.

- **O-Horizon (Organic Layer):** The uppermost layer, composed of dead plant and animal matter in various stages of decomposition. partially decayed organic matter.

- **A-Horizon (Topsoil):** A mineral layer rich with accumulated humus. This is the dark, nutrient-rich layer where most plant roots and soil organisms are found.

- **E Horizon (Eluviated):** A layer of mineral soil that has lost some of its material through leaching (washing out).

- **B-Horizon (Subsoil):** The layer below the topsoil. It contains less organic matter but accumulates clay, iron, and other materials leached from the layers above.

- **C-Horizon (Parent Material):** The unconsolidated material below the subsoil, which is only slightly weathered.

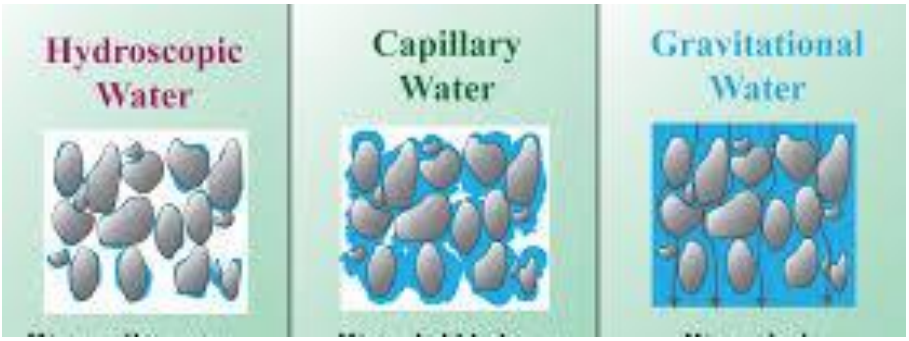
- **R Horizon (Bedrock):** The layer of unweathered, consolidated rock that lies beneath the soil.

14. CLASSIFICATION OF SOIL WATER

(i) Gravitational water: Gravitational water will drain freely from the soil by the force of gravity. gravitational water is of little use to plants water occupies the larger pores resulting poor aeration.

(ii) Capillary water: Capillary water is held in the micropores of soils Capillary water is held so strongly that the force of gravity is not able to separate it f the soil particles. It is the main water available to plants.

(iii) Hygroscopic water: Plants are not able absorb it and thereby unavailable to plants. Some micro-organisms can utilize such form water.



PART-C

15. CONTROL OF WIND EROSION

Vegetative Cover: vegetative cover discourages soil blowing by winds

Mulching: mulching has ability to be effective in controlling wind erosion.

Rough Surface: Wind is more effective on smooth surface than on rough surface.

Organic Matter: This is increasing the stability of the soil structure, for improving water-holding capacity of the soil and for building up its fertility.

Regulating Grazing: Probably the most important cause of wind erosion is overgrazing There must be grazing regulations (prevent over grazing)

Wind Break and Shelterbelt: Trees and shrubs, commonly planted in rows at right angles to the preventing winds are called shelterbelts.

- Control of Water Erosion

There are two methods for the control or management of land eroded by water: (a. Agronomical method, and (b) Mechanical Method.

I. Agronomical Method.

Growing Cover Crops: crops, such as grass, sunhemp, groundnut are excellent soil protection against erosion by water.

Crop Rotation: Growing of two more different crops in sequence in a field for maintaining the soil fertility. Where plants are densely arranged ,it prevent water erosion

Mulching: Mulches of different kinds minimise evaporation and increases absorption of moisture/water. It will reduce soil erosion

Organic Manures: Incorporation of organic manures in the soil improves the soil structure. Granular and crumby structures increase infiltration and permeability in the soil and conserve soil water.

II.Mechanical Methods.

- **Contour Tillage:** On hilly land areas tillage operations and the sowing of crops should be done at right angles manner this will prevent erosion Ploughing is Perpendicular to the slop
- **Pan Breaking and Sub-soiling:** It break the hard pan of the soil and increase infiltration. This increases the water storage capacity of the soil, thus, decreasing run-off and erosion.
- **Water Harvest:** At suitable places like pond and tanks are store rain water.

16. Integrated Nutrient Management (INM)

Integrated Nutrient Management (INM) is an approach to agriculture that combines organic, inorganic, and biological nutrient sources to maintain soil fertility and plant productivity while minimizing environmental impact. This involves using a mix of chemical fertilizers, organic manures (like compost and green manure), and biofertilizers to improve soil health and crop yields sustainably.

Components of Integrated Nutrient Management

The main components of INM are inorganic fertilizers, organic manures, biofertilizers and vermicompost.

1. Inorganic fertilizers: The chemical fertilizers are the main sources of nutrient elements for soils and crops. On applying fertilizers in balanced quantities at proper time by suitable methods.

2. Organic manures: FYM, compost, green manure and oilcakes are the organic manures that supply almost all the essential plant nutrients to the soil.

3. Biofertilizers : Biofertilizers play significant role in improving soil fertility by fixing atmospheric nitrogen. Some important biofertilizers are as follows:

(i) Nitrogen fixing: Rhizobium, azotobacter, azospirillum and blue green algae fix the nitrogen from atmosphere to the soil and increase crop production.

(ii) Phosphate mobilisers: Bacillus pseudomonas and aspergillus.

4. Vermicompost; It is the degradation product of organic wastes by earthworms. The earthworms are effective in converting agricultural waste into compost that is called vermicompost.

Advantages of integrated nutrient management

- Increase soil fertility
- Increase soil productivity
- Improve physical condition of the soil
- Increase the availability of essential nutrients
- Increase the crop yield
- It keeps the soil in good health
- It increases the fertilizer use capacity